IZZ FARIS PUTRA BIN MASKHURIN

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Professional Summary

Computer Engineering undergraduate at SIT with hands-on experience designing, interfacing, and debugging microcontroller-based systems (ESP32, MSP430, Arduino) and digital logic on FPGAs. Proficient in C/C++, hardware communication protocols (UART, SPI, I2C), and modern OOP principles. Proven ability to deploy robust, low-level firmware and edge-AI solutions across collaborative engineering projects.

Education

Singapore Institute of Technology | Computer Engineering — Singapore
Bachelor of Engineering with Honours in Computer Engineering — Sept 2025 – Present
- Expected graduation: May 2028 | Coursework: Embedded System Design, Instrumentation & Data Analytics, Sensors & Control, Database Systems and Object-Oriented Programming

Ngee Ann Polytechnic | Mechanical Engineering — Singapore
Diploma in Mechanical Engineering — Apr 2020 – May 2023

Work Experience

Ministry of Education | Substitute Teacher — Jul 2025 – Sep 2025
- Delivered lessons and managed classrooms in the absence of regular teachers, ensuring continuity in learning.
- Followed lesson plans, maintained discipline, and provided academic guidance.
- Engaged students through interactive methods (e.g., educational games, videos).
- Assisted with student supervision, school events, and light administrative tasks.
- Developed adaptability, classroom management, and ability to connect with diverse age groups.

Full-Time National Serviceman, Fire & Rescue Specialist | Singapore Civil Defence Force — Singapore
May 2023 – May 2025
- Deployed in firefighting and rescue operations, utilizing SCBA gear to combat structural and vehicular fires while executing urban search-and-rescue and emergency civilian evacuation.
- Conducted hands-on training for junior firefighters on essential equipment and techniques, fostering a supportive learning environment.
- Participated in major national events (F1 Standby @ Marina Bay, NDP 2024 TD40), ensuring rapid emergency response readiness and operational excellence.
- Assisted in risk assessments, equipment preparation, and vehicle readiness for high-profile events, maintaining professional standards.

Projects

Automated Computer Vision Attire Check System | Python, OpenCV, YOLO, Raspberry Pi 5, Edge AI, PyQt
SIT CEG1010 Integrated Team Project | Five-person team
- Engineered an edge-AI lab attire detection and safety enforcement system on a Raspberry Pi 5 utilizing a global shutter camera.
- Trained and optimized a custom YOLO model using OpenCV, establishing strict bounding baselines to resolve subjective classification edge cases.
- Integrated real-time edge inference with a PyQt GUI and an LCD screen for dynamic access restriction and visual feedback.

Automated Smart Mailbox & Delivery Node | MSP430, ESP32, Arduino, UART/SPI/I2C
SIT CEG2001/CEG2002 Embedded Systems | In Progress | Five-person team
- Engineering a multi-MCU distributed IoT mailbox system establishing bidirectional UART communication across an MSP430, an ESP32, and an Arduino.
- Integrating optical, weight, and door-state sensors using hardware interrupts and low-power sleep modes on the MSP430 to maximize battery life.
- Leveraging ESP32 Wi-Fi capabilities to publish real-time parcel arrival notifications and status telemetry to a remote dashboard (Telegram).

Driverless Public Transportation & Shuttle Scheduling System | C++, OOP, SOLID Principles
SIT INF1009 Object-Oriented Programming | Five-person team
• Built a modular, layered autonomous shuttle route planner and passenger dispatch system in modern C++ adhering strictly to SOLID principles.
• Implemented Strategy and Factory design patterns to decouple route assignment, vehicle capacity limits, and passenger pickup within strict 10-minute arrival windows.
• Utilized modern C++ smart pointers for safe memory management, alongside robust input validation and file serialization helpers.

FPGA-Based Sequential Logic Traffic Controller | Logic Programming, VHDL, Control Logic, Basys3
SIT CEG1007 Digital Systems | Five-person team
• Designed a deterministic finite state machine (FSM) utilizing sequential logic design principles to simulate and control complex synchronization patterns.
• Developed, synthesized, and rigorously simulated hardware description code (VHDL) using Xilinx Vivado to ensure optimal logic gate utilization and timing accuracy.
• Deployed the optimized bitstream onto a Basys 3 FPGA development board, achieving reliable hardware execution of the control algorithms.

Bus Captain Telematic Risk & Incident Predictive Pipeline | Python, Streamlit, Matplotlib, NumPy, Pandas, Scikit-Learn, Docker
SIT CEG2003 Data Analytics - Machine Learning | In Progress | Five-person team
• Developing an end-to-end machine learning analytics pipeline within containerized Docker environments to identify high-risk driving behaviors among public bus captains.
• Ingesting and preprocessing high-frequency telemetric sensor alarm logs and historical accident datasets using Pandas and NumPy.
• Engineering behavioral features (harsh braking, rapid acceleration, lane swerves) to train classification models capable of predicting incident-prone driver profiles.

Awards & Competitions

Nebula X Hackathon 2026 | 2nd Place | September 2026
LTA @NUS | Five-person team
• Collaborated in a team to build PLiZ, an AI-assisted railway maintenance workspace using FastAPI and React to automate crew scheduling and dynamically resolve track-access conflicts during operational disruptions.
• Emerged as 2nd Place after competing with over 240+ national teams formed across Singapore's autonomous universities.

Agnes AI Hackathon 2026 | 1st Place | June 2026
Agnes AI & SMU AI @ SMU
• Collaborated in a team of 5 to develop FitZ, an AI-powered digital wardrobe and automated outfit planner utilizing Agnes AI APIs to reduce daily decision stress and impulsive buying.
• Emerged as champions against 37+ teams in Singapore, formed across Singapore's autonomous universities and polytechnic

Co-Curricular Activities

Singapore Institute of Technology
Helix First Year Experience (FYE) 2026 Facilitator | May 2026 – Aug 2026
• Guided incoming freshmen through orientation as an Orientation Group Leader. Facilitated team-building initiatives, shared university transition strategies, and fostered peer bonding and engagement across the cohort.

MENDAKI Club
Youth Mentor | Jan 2025 – Aug 2025
• Mentored secondary school students over six months through experiential team-building activities and reflective workshops to foster resilience, personal growth, and community belonging.

Skills and Languages

Programming: Python | C++ | C | SQL
Software Design: Object-Oriented Programming | SOLID | Strategy and Factory patterns
Libraries & Tools: OpenCV | NumPy | Streamlit | Matplotlib | GitHub | Visual Studio | YOLO
Systems & Data: Raspberry Pi | MySQL | NoSQL databases | Electrical diagnostics
Languages: English (Fluent) | Malay (Proficient)